emphasizes the distinction between units of flux and intensity in radiometry and photometry for example. It is noted that in mathematics and some areas of science rad and sr are omitted.

It is especially important to have a clear description of any quantity with the unit one (see section 5.4.7) that is expressed as a ratio of quantities of the same kind (for example length ratios or amount fractions) or as a number of entities (for example number of photons or decays).

2.3.4 Derived units

Derived units are defined as products of powers of the base units. When the numerical factor of this product is one, the derived units are called *coherent derived units*. The base and coherent derived units of the SI form a coherent set, designated the *set of coherent SI units*. The word "coherent" here means that equations between the numerical values of quantities take exactly the same form as the equations between the quantities themselves.

Some of the coherent derived units in the SI are given special names. Table 4 lists 22 SI units with special names. Together with the seven base units (Table 2) they form the core of the set of SI units. All other SI units are combinations of some of these 29 units.

It is important to note that any of the seven base units and 22 SI units with special names can be constructed directly from the seven defining constants. In fact, the units of the seven defining constants include both base and derived units.

The CGPM has adopted a series of prefixes for use in forming the decimal multiples and sub-multiples of the coherent SI units (see chapter 3). They are convenient for expressing the values of quantities that are much larger than or much smaller than the coherent unit. However, when prefixes are used with SI units, the resulting units are no longer coherent, because the prefix introduces a numerical factor other than one. Prefixes may be used with any of the 29 SI units with special names with the exception of the base unit kilogram, which is further explained in chapter 3.

Table 4. The 22 SI units with special names and symbols

Derived quantity	Special name of unit	Symbol	Unit expressed in terms of base units [(a)]	Unit expressed in terms of other SI units
plane angle	radian [(b)]	rad	[(b)]	1
solid angle	steradian [(c)]	sr	[(c)]	1
frequency	hertz [(d)]	Hz	s^{-1}	
force	newton	N	kg m s^{-2}	
pressure, stress	pascal	Pa	kg m^{-1} s^{-2}	N/m^2
energy, work, amount of heat	joule	J	kg m^2 s^{-2}	N m
power, radiant flux	watt	W	kg m^2 s^{-3}	J/s
electric charge	coulomb	C	A s	
electric potential difference [(e)]	volt	V	kg m^2 s^{-3} A^{-1}	W/A
capacitance	farad	F	kg^{-1} m^{-2} s^4 A^2	C/V
electric resistance	ohm	Ω	kg m^2 s^{-3} A^{-2}	V/A
electric conductance	siemens	S	kg^{-1} m^{-2} s^3 A^2	A/V
magnetic flux	weber	Wb	kg m^2 s^{-2} A^{-1}	V s
magnetic flux density	tesla	T	kg s^{-2} A^{-1}	Wb/m^2
inductance	henry	H	kg m^2 s^{-2} A^{-2}	Wb/A

Celsius temperature	degree Celsius [f]	°C	K	
luminous flux	lumen	lm	cd sr [g]	cd sr
illuminance	lux	lx	cd sr m^{-2}	lm/m^2
activity referred to a radionuclide [d, h]	becquerel	Bq	s^{-1}	
absorbed dose, kerma	gray	Gy	$m^2\ s^{-2}$	J/kg
dose equivalent	sievert [i]	Sv	$m^2\ s^{-2}$	J/kg
catalytic activity	katal	kat	mol s^{-1}	

(*a*) The order of symbols for base units in this Table is different from that in the 8th edition following a decision by the CCU at its 21st meeting (2013) to return to the original order in Resolution 12 of the 11th CGPM (1960) in which newton was written kg m s^{-2}, the joule as kg m^2 s^{-2} and J s as kg m^2 s^{-1}. The intention was to reflect the underlying physics of the corresponding quantity equations although for some more complex derived units this may not be possible.

(*b*) The radian is the coherent unit for plane angle. One radian is the angle subtended at the centre of a circle by an arc that is equal in length to the radius. This suggests rad = m/m but this representation is not intrinsic and may be misleading since angle is not the same kind of quantity as other length ratios. An alternative definition is that a right angle is equal to $\pi/2$ rad. The radian is also the coherent unit for phase angle. For periodic phenomena, the phase angle increases by 2π rad in one period.

(*c*) The steradian is the coherent unit for solid angle. One steradian is the solid angle subtended at the centre of a sphere by an area of the surface that is equal to the squared radius. This suggests sr = m^2/m^2, but this representation is not intrinsic and may be misleading since solid angle is not the same kind of quantity as other area ratios. An alternative definition is that a complete sphere subtends 4π sr about its centre.

(*d*) The hertz shall only be used for periodic phenomena and the becquerel shall only be used for stochastic processes in activity referred to a radionuclide.

(*e*) Electric potential difference is also called “voltage” in many countries, as well as “electric tension” or simply “tension” in some countries.

(*f*) The degree Celsius is used to express Celsius temperatures. The numerical value of a temperature difference or temperature interval is the same when expressed in either degrees Celsius or in kelvins.

(*g*) In photometry the name steradian and the symbol sr are usually retained in expressions for units.

(*h*) Activity referred to a radionuclide is sometimes incorrectly called radioactivity.

(*i*) See CIPM Recommendation 2 on the use of the sievert (PV, 2002, **70**, 205).

The seven base units and 22 units with special names and symbols may be used in combination to express the units of other derived quantities. Since the number of quantities is without limit, it is not possible to provide a complete list of derived quantities and derived units. Table 5 lists some examples of derived quantities and the corresponding coherent derived units expressed in terms of base units. In addition, Table 6 lists examples of coherent derived units whose names and symbols also include derived units. The complete set of SI units includes both the coherent set and the multiples and sub-multiples formed by using the SI prefixes.